Subject Description Form

Subject Code	EEE503
Subject Title	Microphotonics and Quantum Photonic Systems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to bring in-depth treatment of fundamental concepts behind light propagation in various photonic components and quantum devices. Focusing on the basic underlying theory of silicon photonics and quantum, and description of different quantum photonic devices and systems. Applications are briefly described.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills a. Understand the waveguides and basic waveguide-based components. b. Understand the basic operation of the most important passive and active photonic components. c. Understand the basic operation of quantum devices and systems. Category B: Attributes for all-roundedness d. Communicate effectively. e. Think critically and creatively. f. Assimilate new technological development in related field
Subject Synopsis/ Indicative Syllabus	**Introduction of microphotonics**: Understanding of the behavior of light propagation through structures with small features of the order of the wavelength of light. **Matrix descriptions of wave propagation**: Transfer matrices and representation of light polarization, Jones matrices. **Thin films**: Reflection and transmission of layered media: transfer matrix method and coatings. **Fourier optics**: Diffraction theory: Fresnel and Fraunhofer, Fourier transform properties of lenses. **Dielectric waveguides**: Theory of slab and stripe waveguide. Waveguide structures: bends, junctions, and couplers. **Photonic components and quantum devices**: Light modulators (electro-optical, thermo-optical), Polarisation components, Optical switching, and Single photodetectors. **Microphotonic measurement and device characterization**: MZI modulator, Fabry-Perot, FTIR, grating.

Teaching/Learning Methodology

The basic concepts of photonic structures and components will be introduced in lectures. The basic numerical analysis methods will be discussed. At the end of this subject, the student will be able to understand the behavior of a variety of photonic structures, components, and systems on the basis of how light propagates through them. The student will also be able to solve wave propagation problems with generic solutions. Students will also be required to study one of the photonic devices and share their findings with other classmates through presentations.

Teaching/Learning Methodology	Intended subject learning outcomes					
	a	b	c	d	e	f
Lectures	✓	✓	✓	✓	✓	✓
Exercises	✓	✓	✓		✓	✓
Case study and presentation	✓	✓	✓	✓	✓	✓

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	f
1. Assignment & Exercises	40%	✓	✓	✓	✓	✓	✓
2. Examination	60%	✓	✓	✓		✓	✓
Total	100 %						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

Assignments: Let students review the taught materials, do further reading for deeper learning and understand better of the taught knowledge. Students will practice the obtained knowledge in the associated exercises.

Examination: The examination will evaluate students' understanding of the microphotonics and quantum photonics

Student Study Effort Expected

Class contact:	
▪ Lectures	30 Hrs.
▪ Tutorials & Case study	9 Hrs.
Other student study effort:	
▪ Reading and self-study	81 Hrs.
Total student study effort	120 Hrs.

Reading List and References	**Textbooks** • Born, Max. Principles of Optics: 60th Anniversary Edition. Seventh (expanded) anniversary edition, 60th anniv. Cambridge University Press, 2019. • Jamroz, Wes R. Applied Microphotonics. United States: CRC Press, 2018. • Hunsperger, Robert G. Integrated Optics Theory and Technology. 6th ed. New York ; Springer, 2009. **References** • Chrostowski, L., & Hochberg, M. E. Silicon photonics design. Cambridge University Press. 2015. • Rigneault, Hervé. Nanophotonics. London ; ISTE, 2006. • Freeman, M. H. (Michael Harold), and C. C. Hull. Optics. 11th ed. London: Butterworth-Heinemann, 2003. • Bass, Michael, Virendra N. Mahajan, Eric W. Van Stryland, Jay M. Enoch, Guifang Li, Carolyn Ann. MacDonald, and Casimer. DeCusatis. Handbook of Optics. 3rd ed. / Michael Bass, Editor-In-Chief. New York: McGraw-Hill, 2010.